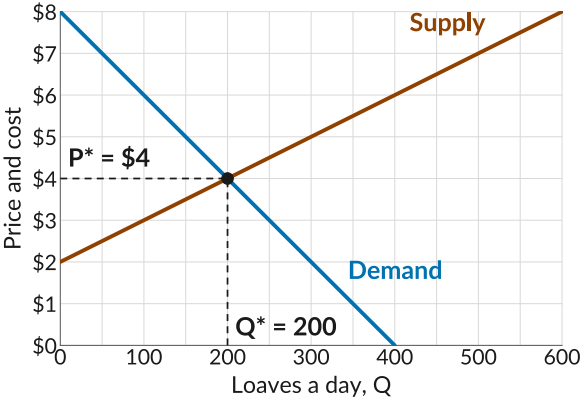
Who Gets What:
Allocation, Distribution, and the Gains From Trade

ECON 201: Principles of Microeconomics
Lecture 17

Div Bhagia

Last Class: The Bread Market

- With many bakeries selling the same loaf, each is a **price-taker**.
- The supply curve is the marginal cost of the Qth loaf, and the demand curve is the Qth buyer's willingness to pay.
- They cross at $P^* = \$4$ and $Q^* = 200$ (WTP = MC).

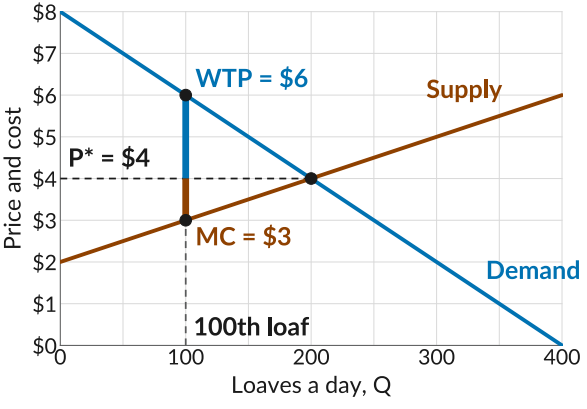


Today: who gains from these trades, how the gains are split, and whether the market leaves any gains on the table.

The Gains From One Loaf

Buyers and bakeries trade because both gain. The **gains from trade** are what each side gets from a trade compared with not trading at all.

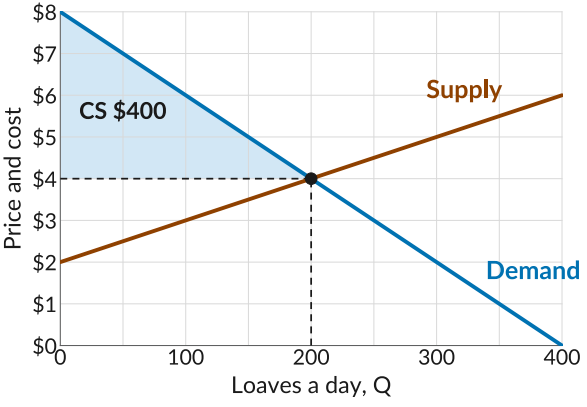
- The 100th buyer would pay up to \$6 but pays \$4, so they gain $6 - 4 = 2$.
- The 100th loaf costs \$3 to bake and sells for \$4, so the bakery gains $4 - 3 = 1$.
- Every buyer who gets a loaf gains WTP $- P$, and every loaf sold earns its bakery $P -$ MC, whoever buys from whom.



Consumer Surplus

Consumer surplus (CS): the sum of WTP $-$ P over every loaf bought.

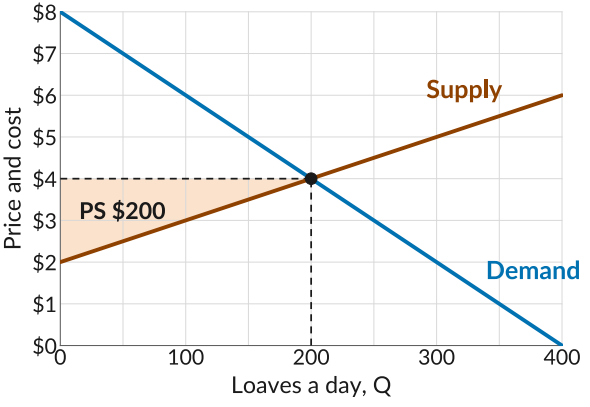
- Every buyer up to the 200th gains the gap between the demand curve and the price.
- Added up:
  $\text{CS} = \frac{1}{2} \times 200 \times (8 - 4) = 400.$



Producer Surplus

Producer surplus (PS): the sum of $P - \text{MC}$ over every loaf sold.

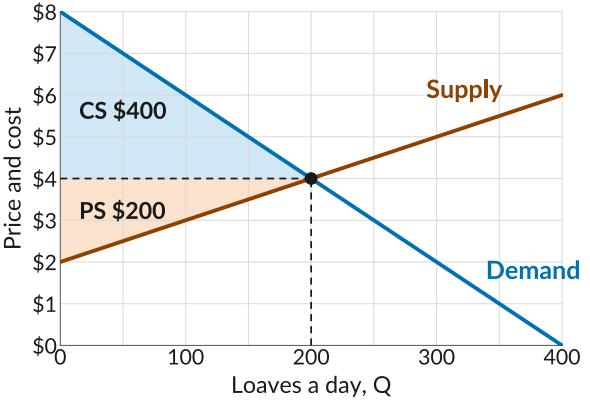
- Every loaf up to the 200th earns its bakery the gap between the price and the supply curve.
- Added up:
  $\text{PS} = \frac{1}{2} \times 200 \times (4 - 2) = 200.$
- Producer surplus is profit plus fixed costs.



Total Surplus

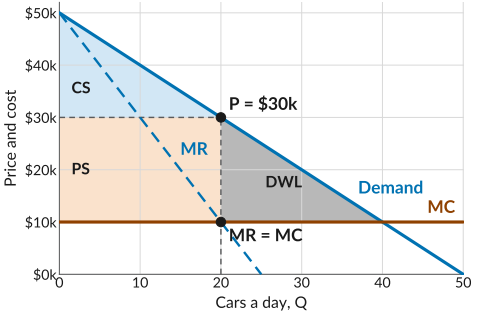
Total surplus: consumer surplus plus producer surplus, the gains from all the trades in the market.

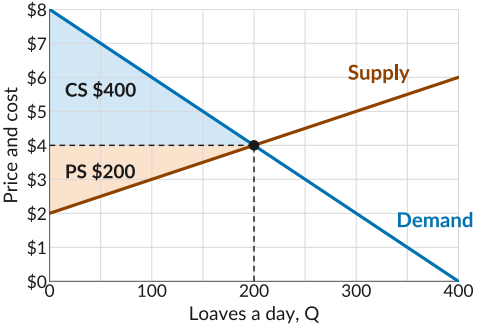
- Total surplus is $400 + 200 = 600$, the whole shaded area.
- Buyers get two thirds of it and bakeries one third.
- Notice any difference from the surplus in the market for Beautiful Cars?



No Deadweight Loss

Beautiful Cars priced above marginal cost, at the quantity where MR = MC, so buyers who valued a car above its cost went without: a **deadweight loss**.



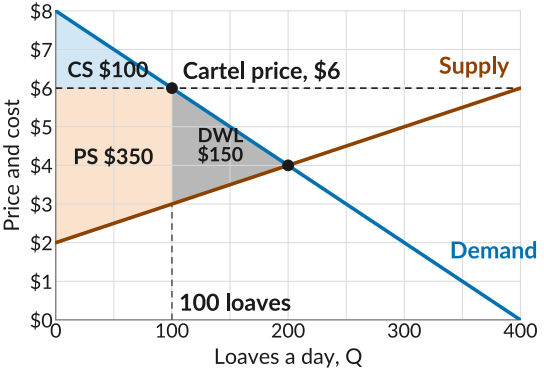


In the bread market, every loaf worth more than it costs gets baked: a competitive equilibrium maximizes the total surplus.

A Bread Cartel

One way to see this: suppose the bakeries form a **cartel**. They agree to charge \$6 and to bake only what buyers want at that price, 100 loaves.

- Buyers: CS $= \frac{1}{2} \times 100 \times 2 = 100$.
- Bakeries:
  PS $= 3 \times 100 + \frac{1}{2} \times 100 \times 1 = 350$.
- The trades between 100 and 200 no longer happen: a deadweight loss of $\frac{1}{2} \times 100 \times 3 = 150$. Total surplus falls to \$450.
- A price held **below** \$4 would leave trades undone too.



Is the Outcome Pareto Efficient?

- An allocation is **Pareto efficient** if no feasible change could make someone better off without making anyone worse off.
- Since all the gains from trade are exhausted, no change in how many loaves are baked, or in who bakes and who buys them, can make a buyer or bakery better off without hurting another.
- So the equilibrium allocation is Pareto efficient, provided nobody outside the market is affected by the trades. If baking were noisy or polluting, the neighbours' costs would have to count too.
- An **external effect** is a cost or benefit that one person's action imposes on others, which the person acting does not take into account.

When Does This Result Hold?

The Pareto efficiency of a competitive equilibrium is often read as a strong argument for markets. It holds only under four conditions.

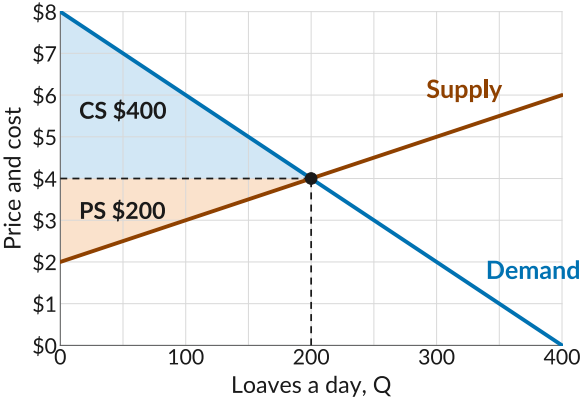
- Many buyers and sellers of identical goods.
- The market is at its equilibrium, with everyone a price-taker.
- Trade in the market has no external effects.
- Each trade is governed by a **complete contract**: one that covers everything either side cares about and can be enforced at almost no cost.

In Practice

- Most firms sell goods that are differentiated in some way, as we saw before the midterm. Even bakeries selling identical loaves differ in location, service, and what else they sell.
- Evidence of price-taking is hard to find.
- Many goods have external effects, such as the carbon emissions from producing them.
- Contracts are sometimes incomplete. The buyer of a used car cannot tell whether it is reliable.

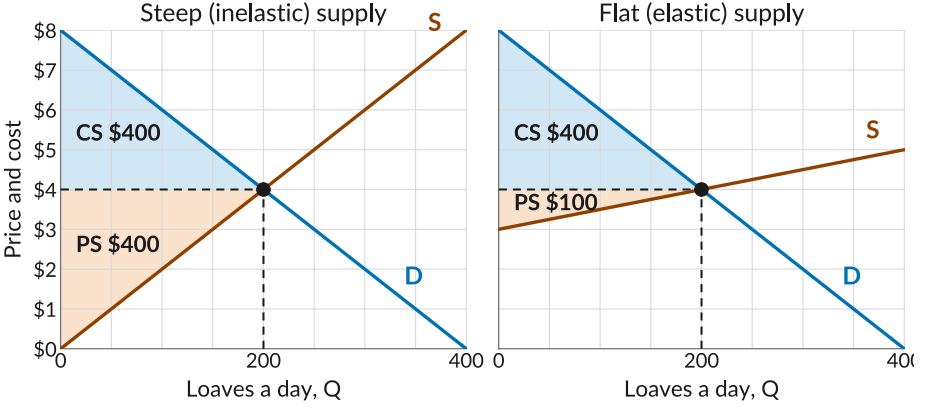
Who Gets the Gains?

- Efficiency is one criterion for judging an allocation. Fairness is another.
- Surpluses say nothing about fairness, but they do show how the gains are split.
- Here buyers get 2/3 and sellers 1/3 of the surplus, because demand is steeper than supply.

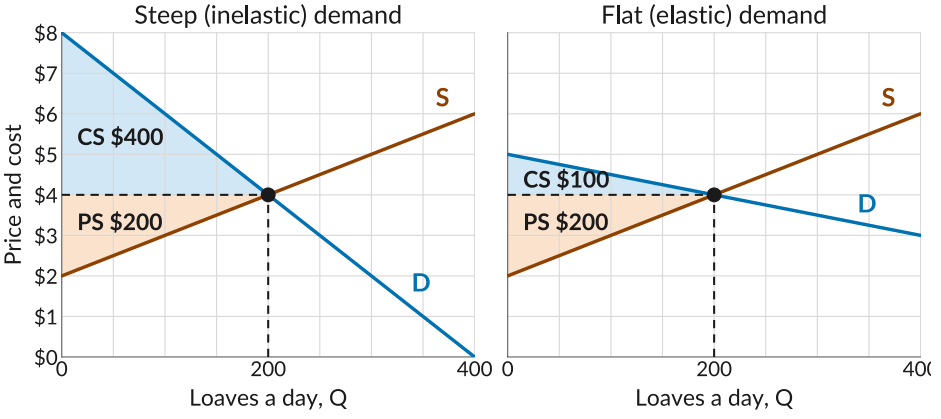


Elasticity of Supply and the Split

The **elasticity of supply** measures how much the quantity supplied responds to the price. A flat supply curve is elastic, a steep one inelastic.



Elasticity of Demand and the Split



Elasticity and the Split

- A buyer's gain on a loaf is their willingness to pay minus the price, and a bakery's is the price minus marginal cost. Each side's surplus adds up these gaps.
- When demand is inelastic, many buyers value bread well above $4 and would pay far more than they do, so buyers gain a lot. When it is elastic, most buyers would pay only a little more than $4, and would walk away if it rose, so they gain little.
- The same goes for supply. Inelastic supply means many loaves cost well below $4 to bake; elastic supply means most loaves cost nearly $4.
- So how the gains are split depends on the relative elasticities of demand and supply: the less elastic side keeps more.